Analysis and Design of Algorithms

CS3230

Properties of Functions

- Exponentials
- Logarithms
- Summations
- Limits

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Exponentials

$$a^{-1} = 1/a$$

$$(a^m)^n = a^{mn}$$

$$a^m a^n = a^{m+n}$$

$$e^x \geq 1 + x$$

Any exponential function with base $a > 1$ grows faster than any polynomial, $n^k = O(a^n)$

Logarithms

- **Binary log:** $\lg n = \log_2 n$
- **Natural log:** $\ln n = \log_e n$
- **Exponentiation:** $\lg^k n = (\lg n)^k$
- **Composition:** $\lg \lg n = \lg(\lg n)$

$$a = b^{\log_b a}$$

$$\log_c(ab) = \log_c a + \log_c b$$

$$\log_b a^n = n \log_b a$$

$$\log_b a = \frac{\log_c a}{\log_c b}$$

$$\log_b(1/a) = -\log_b a$$

$$\log_b a = \frac{1}{\log_a b}$$

$$a^{\log_b c} = c^{\log_b a}$$

Base of logarithm does not matter in asymptotics

$$\lg n = \Theta(\ln n) = \Theta(\log_{10} n)$$

Exponentials of different bases differ by an exponential factor

$$4^n = \boxed{2^n} 2^n$$

Stirling's approximation

$$n! = \sqrt{2\pi n}\left(\frac{n}{e}\right)^n \left(1 + \Theta\left(\frac{1}{n}\right)\right)$$

$$\log(n!) = \Theta(n \lg n)$$

Summations

Arithmetic Series

$$\sum_{k=1}^{n} k = 1 + 2 + 3 + \cdots + n$$

$$= \frac{1}{2} n(n+1) = \Theta(n^2)$$

Geometric Series

$$\sum_{k=1}^{n} x^k = 1 + x + x^2 + \cdots + x^n$$

$$= \frac{x^{n+1} - 1}{x - 1}$$

$$\sum_{k=1}^{\infty} x^k = \frac{1}{1 - x} \text{ when } |x| < 1$$

Harmonic Series

$$H_n = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n}$$

$$= \sum_{k=1}^{n} \frac{1}{k}$$

$$= \ln n + O(1)$$

Telescoping series

For any sequence $a_0, a_1, \ldots, a_n$

$$\sum_{k=0}^{n-1} (a_k - a_{k+1}) = a_0 - a_n$$

Example

$$\sum_{k=1}^{n-1} \frac{1}{k(k+1)} = \sum_{k=1}^{n-1} \left(\frac{1}{k} - \frac{1}{k+1} \right)$$

$$= 1 - \frac{1}{n}$$

Limits

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} < \infty \implies f(n) \in O(g(n))$$

$$0 < \lim_{n\to\infty} \frac{f(n)}{g(n)} \implies f(n) \in \Omega(g(n))$$

$$0 < \lim_{n\to\infty} \frac{f(n)}{g(n)} < \infty \implies f(n) \in \Theta(g(n))$$

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = 0 \implies f(n) \in o(g(n))$$

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = \infty \implies f(n) \in \omega(g(n))$$

L'Hopital's Rule:

$$\lim_{x \to \infty} \frac{f(x)}{g(x)} = \lim_{x \to \infty} \frac{f'(x)}{g'(x)}$$

Example:

$$\lim_{n\to\infty} \frac{n \log n}{n^2}$$

$$= \lim_{n\to\infty} \frac{\log n}{n}$$

$$= \lim_{n\to\infty} \frac{1/n}{1}$$

L'Hopital's rule

$$= \lim_{n\to\infty} \frac{1}{n} = 0$$

$$\Longrightarrow \; n \log n \in o(n^2)$$

Example:

$$\lim_{n \to \infty} \frac{\lg^b n}{n^a} = 0$$

$$\implies \lg^b n = o(n^a)$$